CARLOS MALAGÓN TENORIO

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Portfolio: https://mtcarlos.github.io/MyPortfolio/ | **GitHub:** https://github.com/mtcarlos | **LinkedIn:** www.linkedin.com/in/carlos-malagon-tenorio

PROFESSIONAL SUMMARY

Highly motivated Audiovisual and Multimedia Systems Engineer student with a strong passion for software development, frontend engineering, and UI/UX design. Adept at bridging the gap between complex technical backends and clean, intuitive user interfaces. Experienced in building full-stack applications with Python, Django, and JavaScript, integrating AI APIs, and designing interactive responsive systems. Fluent in English (C1 level).

TECHNICAL & DESIGN SKILLS

Programming & Frameworks: JavaScript, Python, Django, Node.js, Java, HTML5, CSS3, SQL
UI/UX & Design: Figma, Interactive Prototyping, Wireframing, User Research, Information Architecture
Engineering & Systems: Telecom & Networking Protocols, Audiovisual Signal Processing, Media Streaming Systems
Languages: Spanish (Native), English (C1 - Highly proficient)

KEY PROJECTS

VR Code City: Immersive Code Analysis Environment (Bachelor's Thesis / TFG) |

Lead Full-Stack & WebXR Developer | Website

WebXR, A-Frame, Three.js, JavaScript, WebRTC, Networked-Aframe, Node.js, Python (FastAPI), OpenRouter API, GitHub REST API

- Architected and developed a web-based WebXR platform that dynamically maps GitHub repositories into interactive 3D cities using the "Code City" metaphor, converting directory trees into urban blocks and code metrics (LOC) into buildings.
- Implemented low-latency multi-user collaboration, real-time avatar tracking (6DoF), and spatial presence using WebRTC, Socket.IO, and Networked-Aframe.
- Engineered "The Oracle", an in-VR contextual AI agent leveraging Google Gemma LLMs via OpenRouter and Gaze-Based RAG to inspect, explain, and refactor code directly via spatial raycasting.
- Designed a dual-backend architecture: a Node.js/Express server for asset delivery and WebRTC signaling, paired with an asynchronous Python/FastAPI microservice for partial Git cloning and temporal repository analysis (X-Ray Mode & Time Machine).

Magic: The Gathering Social Network & AI Assistant | *Full-Stack Developer & UI Designer* | Platform

Python, Django, OpenAI/Gemini API, JavaScript, CSS3, SQL

- Architected and developed a full-stack social network web application tailored for the Magic: The Gathering gaming community.
- Integrated cutting-edge Generative AI APIs to implement smart custom features (such as automated deck recommendations and strategic analyses).
- Engineered secure user authentication, personal profiles, and robust backend database handling in Django.
- Designed a clean, fully-responsive visual interface from scratch using CSS/JS for implementation, significantly improving complex gaming data readability.

EDUCATION

B.S. in Audiovisual and Multimedia Systems Engineering	2021 - 2026
Universidad Rey Juan Carlos (URJC), Madrid, Spain	Expected Graduation: 2027
Coursework includes: Data Networks, Web Technologies, Interactive Multimedia Systems, Signal Processing, and Image Synthesis.	

WORK EXPERIENCE

Chess Teacher	09/2025 - 12/2025
Playedu, Pinto, Madrid	Madrid, Spain

- Developed and implemented structured chess curriculums tailored to kids aged 6-12, fostering analytical thinking and problem-solving.
- Maintained strong communication and presentation skills, translating abstract game rules into highly engaging, gamified lessons.